Dhruv Dilipkumar Prajapati

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Mechanical Engineering rising senior student seeking a summer and/or fall 2026 co-op/internship blocks.

EDUCATION

Bachelor of Science in Mechanical Engineering, Rochester Institute of Technology, NY, USA, GPA: 3.50 **Aug 2027**

Coursework: Systems Dynamics & Modeling, Circuits, Heat Transfer, Fluid Mechanics, Intro to Acoustics, Dynamics, Material Science, Semiconductor Innovation & Manufacturing, EUV Photolithography Technology, Thermodynamics, Statics, Strength of Materials, Engineering Measurements lab, Computer Aided Design/Manufacturing, Multivariable Calculus, Ordinary Differential Equations, Boundary-value Problems.

PROFESSIONAL EXPERIENCE

PG&E – Pacific Gas & Electric Company **June 2026 – Sept 2026**
Intern – Failure Analysis Engineer, Simulation & Test **San Ramon, CA**
(ATS Materials & Failure Analysis Group, Wildfire Mitigation)

- Incoming intern supporting failure analysis investigations involving fatigue/fracture assessment, laboratory testing, and FEA simulations.
- Assisting with finite element analysis (FEA) simulations in Ansys to evaluate stress distributions, structural integrity, and potential failure locations.
- Working with engineering teams on microscopy-based inspections and failure investigations, including SEM-assisted fracture surface analysis workflows.
- Utilizing MATLAB for engineering data processing, visualization, and analysis of experimental test results & contributing to technical reports, documentation, and root-cause analysis for reliability and asset integrity assessments

MKS Instruments **Aug 2025 – Dec 2025**
Intern – Mechanical Test Engineer **Rochester, NY**
(Power Solutions Division)

- Designed and analyzed thermal-fluid systems for semiconductor heat sink applications using SolidWorks Flow Simulation (CFD) for new product innovation.
- Correlated simulation results with experimental data (pressure drop, flow rate, velocity) to improve model accuracy and inform design decisions & develop and execute firsthand testing setups using flow meters, pressure gauges, and chillers to validate system performance.
- Improved test efficiency by 30% and reduced cost by 15% through optimization of flow analysis and validation procedures.
- Supported requalification of technical instruments & workstations using SolidWorks CAD design and updated vital information on 2D layouts using ACAD by validating Bill of Materials (BOM) with modified designs.
- Assisted in instrumentation and test system setup (LabVIEW, ultrasonic flowmeters) for hardware validation and burn-in testing involving HVAC controls, bulk liquid nitrogen systems, and environmental ALT HASS/ESS chambers for reliability testing.

RESEARCH EXPERIENCE

Fluids Acoustics STructures Laboratory (FAST Lab) **February 2026 – April 2026**
Undergraduate Researcher **Rochester, NY**

- Designed and fabricated custom mounts and test fixtures using SolidWorks and 3D printing to support experimental setups.
- Developed actuator control systems using Arduino and Pololu Jrk G2 and performed data acquisition and analysis for flow-driven thermal applications.
- Generated and visualized primary and secondary vortex rings, capturing leapfrogging behavior to study vortex dynamics and flow structures.
- Investigated vortex-induced flow and impingement heat transfer in confined geometries using PIV-based experimental methods.

PROJECTS

Quarter-car suspension model (MATLAB/Simulink), Liquid-cooled Heat Sink Analysis, Reversed Engineering Project (SolidWorks + Simulation), Thermocouples, Encoder, Unknown RLC Circuit Blackbox Analysis

CORE COMPETENCIES

- ***Thermal & Fluid Systems***: Heat Transfer, Fluid Mechanics, CFD (SolidWorks Flow Simulation)
- ***CAD Design & Analysis***: SolidWorks, ACAD 2D/3D, FEA (Ansys Mechanical), CATIA v5, GD&T
- ***Programming & Modeling****:* Python, MATLAB, Simulink, CFTool, LabVIEW, Arduino IDE, Pololu jrk G2 microcontroller
- ***Tools & Fabrication***: Lathe, CNC Milling, turning, Surface Finish, Shop Machine tools, Prusa i3 3D Printing machine, Instron machine, Quasar 10 Tensile Tester, Rockwell Tester, Oscilloscopes (Tektronix/Agilent Tech), Keysight Waveform generator, PIV

AWARDS & ACHIEVEMENTS

Dean's List 2025-26 | RIT Founder's Merit-Based Scholarship Award - $92,000 | PMI – Six Sigma Foundations Certificate | MKS Co-op Program Best Presenter Award